CASE STUDY / 2026

Growlytic Marketing Intelligence

A production AI marketing platform for competitor research, creative generation, campaign planning, live ad analytics, and reporting.

Next.js 16 | Prisma / Neon | Anthropic | Google Gemini | Meta & Google Ads | Paddle

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A technical walkthrough of the problem, architecture, implementation, evidence, and next stage.

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Evidence standard

The case study is based on implemented routes, database schema, migrations, production handoff records, and the live public site, which returned HTTP 200 during verification on 20 October 2026. Sensitive credentials, customer records, and billing details are excluded.

SECTION 01

Project Overview

One workspace moves from real competitor ads to AI analysis, generated creative, campaign plans, live account metrics, and client-ready reports.

Growlytic is a production marketing-intelligence SaaS that combines competitor ad discovery, watchlists, swipe boards, AI analysis, creative briefs, copy and image generation, audience segments, campaign planning, ad-account connections, live metrics, comparative performance reports, report chat, PDF export, credits, plans, and administration.

The product connects real advertising and search-data sources to a governed application layer. Next.js server routes isolate credentials and external APIs, Prisma and Neon retain user and analytical state, and AI models are used for interpretation and content generation while deterministic code owns calculations, permissions, credits, and report configuration.

14	81	30	25
major product feature areas	server API routes	database migrations	Prisma data models

Real data over mock dashboards

Competitor ads, account metrics, conversions, and report inputs come from connected or licensed sources.

AI inside guardrails

Models explain and generate; deterministic code controls arithmetic, validation, budgets, permissions, and credit charging.

Safe activation

Generated ads are pushed as paused drafts so the user remains responsible for review and publication.

Measured monetization

Plan limits, brand slots, credit spends, ledger records, refills, Paddle events, and admin operations share one source of truth.

SECTION 02

Platform Architecture

A Next.js application coordinates authentication, persistence, third-party marketing APIs, AI services, background schedules, exports, and production billing.



Layer	Implemented responsibility
Experience	Next.js 16, React 19, TypeScript, Tailwind, protected dashboard routes, onboarding, admin, settings, billing, and responsive report interfaces.
Data	Prisma 7 with Neon PostgreSQL stores users, competitors, ads, analyses, creatives, boards, briefs, reports, connections, credits, plans, and billing state.
Intelligence	Anthropic generates grounded analysis and copy; Google Gemini supports image generation and multimodal interpretation with timeout controls.
Marketing integrations	Meta, Google Ads, SearchAPI, TikTok transparency, account OAuth, conversions, metrics, draft push, and live report data connect external platforms.
Operations	Vercel deployment, cron schedules, Resend email, Vercel Blob, Paddle webhooks, Puppeteer PDF generation, and server-side validation support production use.

Boundary of authority

The model can describe performance and propose creative, but it cannot silently launch ads, invent spend arithmetic, mutate credits, or bypass account permissions. Those actions remain deterministic and auditable.

SECTION 03

Product & Data Model

The database connects competitive intelligence, creative production, owned-account reporting, user entitlements, and billing without collapsing them into one generic record.

Competitive intelligence

Competitors, source ads, AI analyses, watchlists, boards, and board items preserve discovery, evidence, and researcher notes.

Creative workflow

Briefs, generated creatives, audience segments, and pushed-ad records track movement from insight to paused advertising draft.

Planning and reporting

Campaign plans, live metrics, comparative reports, templates, style references, schedules, and PDF exports serve recurring client work.

Accounts and economics

Users, profiles, OAuth accounts, connected ad accounts, credits, plans, payment claims, and Paddle transactions govern access and usage.

Flow	Persisted state	Control
Research	Competitor, ads, analysis, board	Source identity and user ownership
Generate	Creative, image, brief, segment	AI schema validation and one-credit spend
Activate	Connected account, pushed ad	OAuth scope and paused-draft default
Measure	Metrics, conversion selection, report	Account/date-range contracts and deterministic totals
Monetize	Plan, balance, ledger, transaction	Idempotent webhooks and auditable entitlement changes

Thirty schema migrations show the product evolving from basic competitor tracking into a multi-module SaaS while preserving upgrade paths for existing users and production data.

SECTION 04

The Engineering Challenge

External ad platforms, AI providers, OAuth flows, expiring media, billing events, and user-generated reports all fail in different ways.

The product must present a simple workflow even though its dependencies have incompatible schemas, permissions, quotas, regional coverage, asynchronous jobs, expiring URLs, and production-only behavior. Reliability therefore depends on defensive integration design and live verification.

API coverage

Meta, Google, and TikTok expose different search scopes, account permissions, and regional visibility.

Expiring media

Signed ad-image URLs can expire after the browser caches them, breaking server-side generation and saved creative references.

Grounded AI

Analyses must cite real ad evidence and calculations while timeouts prevent one dead media URL from hanging a full request.

OAuth complexity

Account connection requires multi-step provider consent, account selection, token handling, and durable connection state.

Report flexibility

Users compare multiple periods, platforms, conversions, layouts, scorecards, styles, templates, schedules, chat edits, and PDF output.

Billing reliability

Credits, plans, refills, checkout events, refunds, and entitlement changes must remain idempotent across webhook retries.

Production lesson Many critical issues appeared only through live workflows: OAuth ordering, serverless browser packaging, stale sessions, expiring media, platform coverage, webhook events, and production callback behavior.

SECTION 05

Implementation Approach

The product grew through complete vertical slices: source integration, persistent state, user workflow, verification, deployment, and operational follow-up.

1. Establish identity

Implement credentials and Google sign-in, verification, password recovery, dashboard protection, profiles, onboarding, and admin access.

2. Connect research data

Integrate Meta, Google, and supporting search providers; persist competitors, ads, source evidence, and analyses.

3. Build creative loop

Generate copy and images, refine assets, organize boards and briefs, save audiences, and push paused platform drafts.

4. Add planning

Create deterministic campaign tiers, channel budgets, strategies, keyword research, and client-ready PDF exports.

5. Measure performance

Connect ad accounts, fetch live metrics and conversions, compare periods, generate narratives, chat with reports, and schedule delivery.

6. Operate the SaaS

Enforce plans and credits, process Paddle billing, send email, run cron jobs, manage storage, audit production data, and deploy on Vercel.

Deterministic core AI output never owns budget arithmetic, conversion totals, entitlement state, or credit balances. Structured schemas validate model responses while application code retains authoritative business calculations.

- Server-side URL retrieval includes SSRF protections that reject localhost and cloud metadata endpoints.
- Report generation supports saved templates, scorecard selection, drag-and-resize layout, style references, period changes, and PDF export.
- Generated creatives reach Meta or Google as paused drafts, providing an explicit human review checkpoint before spend begins.

SECTION 06

Quality & Verification

Verification spans static checks, real databases, live provider calls, browser-driven production flows, and persisted result cleanup.

Evidence	Verified result	Why it matters
Application scale	40 pages; 81 API routes	The system covers public, authentication, dashboard, admin, integration, reporting, and billing surfaces.
Data evolution	25 models; 30 migrations	Competitive, creative, reporting, account, credit, and billing state is versioned in PostgreSQL.
Build quality	Typecheck, lint, and production build clean	Framework, TypeScript, and deployment packaging errors are caught before release.
Live integration	Real provider and production workflows exercised	Ad data, analysis, OAuth, metrics, drafts, reports, billing, and email were verified beyond mocks.
Availability	Public site returned HTTP 200	The production landing page was reachable on Vercel on 20 October 2026.

Current boundary

External data coverage, ad-platform permissions, quotas, provider review status, and paid API availability remain outside application control. The case study describes implemented workflows and verified examples, not guaranteed access to every brand, account, region, or ad at all times.

- Sensitive credentials and customer account data remain outside documentation and source-controlled case-study assets.
- The system records and surfaces provider-specific empty, unauthorized, or degraded states instead of replacing them with plausible mock results.
- Scheduled reports and billing webhooks use idempotent records so retries do not duplicate delivery or entitlement changes.

SECTION 07

Outcome & Next Stage

Growlytic connects competitive research, content creation, campaign planning, and performance measurement in one production workflow.

Shorter insight loop	Safer execution	Repeatable reporting
Real competitor ads become structured analysis and reusable boards.	Generated assets move to paused drafts with human control.	Live metrics, templates, schedules, chat, and PDF support recurring work.

The platform's value comes from continuity: users can discover market activity, preserve evidence, generate campaigns, connect owned accounts, compare performance, and communicate results without moving through disconnected tools. Technical guardrails keep that continuity from turning into uncontrolled automation.

NEXT STAGE

- Expand observability for provider latency, quota pressure, webhook replay, scheduled-report delivery, and data freshness.
- Continue replacing expiring third-party media references with durable, policy-compliant storage where permitted.
- Strengthen automated end-to-end coverage for OAuth, report generation, draft push, and billing recovery scenarios.

LET'S TALK

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